

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE316	OPERATING SYSTEMS	3	0	0	3	
Course Weightage	ATT: 5 CA: 25 MTT: 20 ETT: 50	Exam Category: 11: Mid Term Exam: All MCQ – End Term Exam: All MCQ				
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT					

Course Outcomes :Through this course students should be able to

CO1 :: recall OS structure, functions, evolution, and process management.

CO2 :: apply CPU scheduling algorithms to optimize process execution.

CO3 :: analyze synchronization mechanisms and classical problems for efficient multi-threaded execution.

CO4 :: apply deadlock handling and security methods for a threat free system.

CO5 :: explain memory management techniques for efficient address space management.

CO6 :: classify file management, device management and inter-process communication methods to facilitate process coordination.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	OPERATING SYSTEM CONCEPTS	ABRAHAM SILBERSCHATZ, PETER B. GALVIN, GERG GAGNE	WILEY

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	DESIGN OF THE UNIX OPERATING SYSTEM	MAURICE J. BACH	Pearson Education India
R-2	REAL-TIME SYSTEMS	JANE W. S. LIU	Pearson Education India

Other Reading (OR)	
Sr No	Journals articles as Compulsary reading (specific articles, complete reference)
OR-1	https://www.udacity.com/course/introduction-to-operating-systems--ud923 ,
OR-2	The Linux scheduler: a decade of wasted cores:: https://dl.acm.org/doi/10.1145/2901318.2901326 ,
OR-3	Persistent Memcached: Bringing Legacy Code to Byte-Addressable Persistent Memory:: https://www.usenix.org/system/files/conference/hotstorage17/hotstorage17-paper-marathe-060717.pdf ,

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OR-4	All File Systems Are Not Created Equal:: https://research.cs.wisc.edu/adsl/Publications/alice-osdi14.pdf ,
OR-5	The Evolution of the Unix Time-sharing System:: http://www.read.seas.harvard.edu/~kohler/class/aosref/ritchie84evolution.pdf ,
OR-6	Disk and File System Analysis:: https://booksite.elsevier.com/samplechapters/9781597495868/Chapter_3.pdf ,
OR-7	https://dextutor.com/courses/operating-systems/ ,

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://nptel.ac.in/courses/106106144	Introduction to Operating Systems, IIT Madras
RW-2	https://nptel.ac.in/courses/106108101	Introduction to Operating System
RW-3	https://nptel.ac.in/courses/106102132	Operating Systems, IIT Delhi
RW-4	https://www.cs.miami.edu/home/geoff/Courses/CSC521-04F/Content/	Operating Systems: Course Content
RW-5	https://www.udacity.com/course/introduction-to-operating-systems--ud923	Introduction to Operating Systems
RW-6	https://learn.saylor.org/course/view.php?id=94	Operating Systems

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Introduction to Operating System(Operating System Meaning)	T-1	RW-1 RW-2 RW-3	Discuss Lecture #0 to give general idea of operating system and its working	Student will come to know how operating system came into existence and how they have evolved over time and its functions	Peer to peer discussion	Linux/Unix/Windows

Week 1	Lecture 1	Introduction to Operating System(OS structure)	T-1	RW-1 RW-2 RW-3	Discuss Lecture #0 to give general idea of operating system and its working	Student will come to know how operating system came into existence and how they have evolved over time and its functions	Peer to peer discussion	Linux/Unix/Windows
		Introduction to Operating System(OS structure)	T-1	RW-1 RW-2 RW-3	Discuss Lecture #0 to give general idea of operating system and its working	Student will come to know how operating system came into existence and how they have evolved over time and its functions	Peer to peer discussion	Linux/Unix/Windows
		Introduction to Operating System(functions of OS)	T-1	RW-1 RW-2 RW-3	Discuss Lecture #0 to give general idea of operating system and its working	Student will come to know how operating system came into existence and how they have evolved over time and its functions	Peer to peer discussion	Linux/Unix/Windows
	Lecture 2	Introduction to Operating System(Multiprogramming and Multiprocessing System)	T-1 R-1 R-2		In this lecture multiprogramming Multiprocessing System will be taught to the system.	Students will learn about In this lecture multiprogramming Multiprocessing System.	Peer to peer discussion and learning by problem solving.	
	Lecture 3	Introduction to Operating System(system calls)	T-1	OR-1 OR-5	To teach about the evolution of operating systems and various system calls	Students will learn about the evolution of operating systems and various system calls.	Peer to peer discussing and live demonstration.	
		Introduction to Operating System(evolution of OSs)	T-1	OR-1 OR-5	To teach about the evolution of operating systems and various system calls	Students will learn about the evolution of operating systems and various system calls.	Peer to peer discussing and live demonstration.	
Week 2	Lecture 4	Process Management(PCB, Operations on Processes)	T-1 R-1	OR-2 RW-4 RW-5 RW-6	To teach fundamentals of process in Unix or Linux environment.	Students will learn the fundamentals of process and PCB.	Peer to peer discussion.	
		Process Management (Process states)	T-1 R-1	OR-2 RW-4 RW-5 RW-6	To teach fundamentals of process in Unix or Linux environment.	Students will learn the fundamentals of process and PCB.	Peer to peer discussion.	

Week 2	Lecture 5	Process Management (Operations on processes)	T-1 R-1		To teach the students various operations on processes and process management.	The students will get to know various operations on processes and process management.	Peer to peer discussion.	
	Lecture 6	Process Management(Co-operating and Independent Processes)	T-1	RW-1 RW-2 RW-3	To describe the fundamental of Inter-process communication and threads concepts	The students will learn Inter-process communication and threads concepts	Peer to peer discussion, live demonstration.	
Week 3	Lecture 7	CPU Scheduling(Types of Scheduling)	T-1 R-2	RW-1 RW-2 RW-3 RW-5	In this lecture fundamentals of CPU scheduling and it's types will be discussed.	The students will learn the basics of CPU scheduling and it's types.	Peer to peer discussion	
		CPU Scheduling(CPU scheduler - preemptive and non preemptive)	T-1 R-2	RW-1 RW-2 RW-3 RW-5	In this lecture fundamentals of CPU scheduling and it's types will be discussed.	The students will learn the basics of CPU scheduling and it's types.	Peer to peer discussion	
	Lecture 8	CPU Scheduling(Scheduling Algorithms)	T-1		In this lecture various scheduling criteria, dispatcher and FCFS algorithm will be discusses.	The students will learn various scheduling criteria, dispatcher, and FCFS scheduling algorithm.	Peer to peer discussion and learning by problem solving.	
		CPU Scheduling(Scheduling criteria)	T-1		In this lecture various scheduling criteria, dispatcher and FCFS algorithm will be discusses.	The students will learn various scheduling criteria, dispatcher, and FCFS scheduling algorithm.	Peer to peer discussion and learning by problem solving.	
		CPU Scheduling (Dispatcher)	T-1		In this lecture various scheduling criteria, dispatcher and FCFS algorithm will be discusses.	The students will learn various scheduling criteria, dispatcher, and FCFS scheduling algorithm.	Peer to peer discussion and learning by problem solving.	
		CPU Scheduling(First come first serve)	T-1		In this lecture various scheduling criteria, dispatcher and FCFS algorithm will be discusses.	The students will learn various scheduling criteria, dispatcher, and FCFS scheduling algorithm.	Peer to peer discussion and learning by problem solving.	
	Lecture 9	CPU Scheduling(Shortest job first)	T-1		In this lecture shortest job first and Round Robin scheduling will be discussed.	The students will learn SJF and Round Robin scheduling.	Learning by problem solving.	
		CPU Scheduling(Round robin)	T-1		In this lecture shortest job first and Round Robin scheduling will be discussed.	The students will learn SJF and Round Robin scheduling.	Learning by problem solving.	

Week 4	Lecture 10	CPU Scheduling(Priority)	T-1		In this lecture priority scheduling and multi-level feedback queue will be discussed.	The student will learn priority scheduling and multi-level feedback queue will be discussed.	Peer to peer discussing and learning by problem solving.	
		CPU Scheduling(Multi level feedback queue)	T-1		In this lecture priority scheduling and multi-level feedback queue will be discussed.	The student will learn priority scheduling and multi-level feedback queue will be discussed.	Peer to peer discussing and learning by problem solving.	
	Lecture 11	CPU Scheduling (multiprocessor scheduling, real time scheduling)	T-1		In this lecture multiprocessor scheduling, real time scheduling and thread scheduling will be discussed.	The students will learn multiprocessor scheduling, real time scheduling and thread scheduling.	Peer to peer discussion and learning by problem solving.	
		CPU Scheduling(thread scheduling)	T-1		In this lecture multiprocessor scheduling, real time scheduling and thread scheduling will be discussed.	The students will learn multiprocessor scheduling, real time scheduling and thread scheduling.	Peer to peer discussion and learning by problem solving.	
	Lecture 12	Process Synchronization (Critical Section Problem)	T-1		In this lecture co-operating processes will be introduced and fundamental of critical section problem will be discussed	The student will get to know about co-operating processes and critical section problem.	Peer to peer discussion	
		Process Synchronization (Co-operating processes)	T-1		In this lecture co-operating processes will be introduced and fundamental of critical section problem will be discussed	The student will get to know about co-operating processes and critical section problem.	Peer to peer discussion	
Week 5	Lecture 13	Process Synchronization (Concurrent processes)	T-1		In this lecture the teacher will discuss the notion of concurrent processes and process hierarchy. Teacher can discuss fork-join and cobegin-coend construct of concurrent processing.	The students will get to know about concurrent processes and process heirarchy.	peer to peer discussion and live demonstration using Linux.	

Week 5	Lecture 13	Process Synchronization (Hierarchy of processes)	T-1		In this lecture the teacher will discuss the notion of concurrent processes and process hierarchy. Teacher can discuss fork-join and cobegin-coend construct of concurrent processing.	The students will get to know about concurrent processes and process heirarchy.	peer to peer discussion and live demonstration using Linux.	
	Lecture 14	Process Synchronization (Semaphores)	T-1		In this lecture precedence graphs and synchronization primitives' like semaphore mutex will be discussed.	The student will learn creation of precedence graphs and uses of synchronization constructs.	Peer to peer discussion/ learning by problem solving /live demonstration	
		Process Synchronization (Precedence graph)	T-1		In this lecture precedence graphs and synchronization primitives' like semaphore mutex will be discussed.	The student will learn creation of precedence graphs and uses of synchronization constructs.	Peer to peer discussion/ learning by problem solving /live demonstration	
	Lecture 15	Process Synchronization (Producer consumer problem)	T-1		In this lecture classical synchronization problems will be discussed.	The student will get to know about classical synchronization problem like producer-consumer problem.	Peer to peer discussion.	
		Process Synchronization (classical two process and n-process solutions)	T-1		In this lecture classical synchronization problems will be discussed.	The student will get to know about classical synchronization problem like producer-consumer problem.	Peer to peer discussion.	
Week 6	Lecture 16				Test			
	Lecture 17	Process Synchronization (Monitors)	T-1		In this lecture Dining Philosophers problem, and reader-writer problem will be discussed along with monitors and hardware primitives for synchronization.	The students will learn about Dining Philosopher Problem and Reader-writer Problem and their probable solutions using hardware primitives.	Peer to peer discussion and learning by problem solving.	

Week 6	Lecture 17	Process Synchronization (Dining Philosopher Problem)	T-1		In this lecture Dining Philosophers problem, and reader-writer problem will be discussed along with monitors and hardware primitives for synchronization.	The students will learn about Dining Philosopher Problem and Reader-writer Problem and their probable solutions using hardware primitives.	Peer to peer discussion and learning by problem solving.	
		Process Synchronization (Reader-writer Problem)	T-1		In this lecture Dining Philosophers problem, and reader-writer problem will be discussed along with monitors and hardware primitives for synchronization.	The students will learn about Dining Philosopher Problem and Reader-writer Problem and their probable solutions using hardware primitives.	Peer to peer discussion and learning by problem solving.	
		Process Synchronization (hardware primitives for synchronization)	T-1		In this lecture Dining Philosophers problem, and reader-writer problem will be discussed along with monitors and hardware primitives for synchronization.	The students will learn about Dining Philosopher Problem and Reader-writer Problem and their probable solutions using hardware primitives.	Peer to peer discussion and learning by problem solving.	
	Lecture 18	Threads(Overview)	T-1		In this lecture Threads concept and multithreading models will be discussed.	The students will learn Threads concept and multithreading models will be discussed.	peer to peer discussing	
		Threads(Multithreading Models)	T-1		In this lecture Threads concept and multithreading models will be discussed.	The students will learn Threads concept and multithreading models will be discussed.	peer to peer discussing	
Week 7	Lecture 19	Threads(scheduler activations)	T-1		In this lecture few programming examples to create multithreaded process will be discussed.	The students will learn creation of multithreaded processes.	Live demonstration.	
		Threads(examples of threaded programs)	T-1		In this lecture few programming examples to create multithreaded process will be discussed.	The students will learn creation of multithreaded processes.	Live demonstration.	

		SPILL OVER						
Week 7	Lecture 20				Spill Over			
	Lecture 21				Spill Over			
		MID-TERM						
Week 8	Lecture 22	Deadlock(Deadlock Characterization)	T-1	OR-7	In this lecture Deadlock and essential conditions of deadlock occurrence will be discussed.	The students will get to know the deadlock and its essential conditions.	Peer to peer discussion	
	Lecture 23	Deadlock(Deadlock Avoidance & Detection)	T-1	OR-7	In this lecture deadlock avoidance algorithms will be discussed.	The students will learn deadlock avoidance will be discussed	peer to peer discussion and learning by problem solving.	
	Lecture 24	Deadlock(Handling of deadlocks- Deadlock Prevention)	T-1		In this lecture Deadlock Prevention methods will be discussed.	The students will learn Deadlock Prevention methods will be discussed.	Peer to peer discussion.	
Week 9	Lecture 25	Deadlock(Deadlock Recovery)	T-1		In this lecture Deadlock recovery will be discussed.	The students will learn various ways of Deadlock recovery.	peer to peer discussion	
		Deadlock(Starvation)	T-1		In this lecture Deadlock recovery will be discussed.	The students will learn various ways of Deadlock recovery.	peer to peer discussion	
	Lecture 26	Protection and Security (Need for Security)	T-1		In this lecture need of security and various types of security threats will be discussed.	The students will get to know about need of security and various types of security threats	Peer to peer discussion.	
		Protection and Security (Security Vulnerability like Buffer overflow)	T-1		In this lecture need of security and various types of security threats will be discussed.	The students will get to know about need of security and various types of security threats	Peer to peer discussion.	
		Protection and Security (Trapdoors, Backdoors, cache poisoning etc)	T-1		In this lecture need of security and various types of security threats will be discussed.	The students will get to know about need of security and various types of security threats	Peer to peer discussion.	
		Protection and Security (Authentication-Password based Authentication)	T-1		In this lecture need of security and various types of security threats will be discussed.	The students will get to know about need of security and various types of security threats	Peer to peer discussion.	

Week 9	Lecture 26	Protection and Security (Password Maintenance & Secure Communication)	T-1		In this lecture need of security and various types of security threats will be discussed.	The students will get to know about need of security and various types of security threats	Peer to peer discussion.	
		Protection and Security (Application Security - Virus, Program Threats)	T-1		In this lecture need of security and various types of security threats will be discussed.	The students will get to know about need of security and various types of security threats	Peer to peer discussion.	
	Lecture 27	Protection and Security (Goals of protection)	T-1		Various protection mechanism will be discussed.	The students will learn protection mechanism.	peer to peer discussion.	
		Protection and Security (Principles of protection)	T-1		Various protection mechanism will be discussed.	The students will learn protection mechanism.	peer to peer discussion.	
		Protection and Security (Domain of protection)	T-1		Various protection mechanism will be discussed.	The students will learn protection mechanism.	peer to peer discussion.	
		Protection and Security (Access matrix)	T-1		Various protection mechanism will be discussed.	The students will learn protection mechanism.	peer to peer discussion.	
		Protection and Security (Implementation of access matrix)	T-1		Various protection mechanism will be discussed.	The students will learn protection mechanism.	peer to peer discussion.	
		Protection and Security (System and network threats)	T-1		Various protection mechanism will be discussed.	The students will learn protection mechanism.	peer to peer discussion.	
		Protection and Security (Examples of attacks)	T-1		Various protection mechanism will be discussed.	The students will learn protection mechanism.	peer to peer discussion.	
Week 10	Lecture 28	Memory Management (Logical & Physical Address Space)	T-1		In this lecture fundamentals of memory management and contiguous memory allocation techniques will be discussed.	The students will learn fundamentals of memory management and contiguous memory allocation techniques.	Peer to peer discussion and learning by problem solving.	
		Memory Management (Contiguous Memory allocation)	T-1		In this lecture fundamentals of memory management and contiguous memory allocation techniques will be discussed.	The students will learn fundamentals of memory management and contiguous memory allocation techniques.	Peer to peer discussion and learning by problem solving.	
	Lecture 29				Project			

Week 10	Lecture 30	Memory Management (Paging)	T-1 R-2	OR-7	In this lecture fundamentals of paging, multi level paging and segmentation will be discussed.	The students will get to know fundamentals of paging, multi level paging and segmentation	Peer to peer discussion.	
		Memory Management (Segmentation)	T-1 R-2	OR-7	In this lecture fundamentals of paging, multi level paging and segmentation will be discussed.	The students will get to know fundamentals of paging, multi level paging and segmentation	Peer to peer discussion.	
		Memory Management (Schemes - Paging - simple and multi level)	T-1 R-2	OR-7	In this lecture fundamentals of paging, multi level paging and segmentation will be discussed.	The students will get to know fundamentals of paging, multi level paging and segmentation	Peer to peer discussion.	
Week 11	Lecture 31	Memory Management (Swapping)	T-1		In this lecture the concept of virtual memory and demand paging will be discussed.	The students will learn the concept of virtual memory and demand paging.	Peer to peer discussing.	
		Memory Management(Page interrupt fault)	T-1		In this lecture the concept of virtual memory and demand paging will be discussed.	The students will learn the concept of virtual memory and demand paging.	Peer to peer discussing.	
		Memory Management (Overlays - swapping)	T-1		In this lecture the concept of virtual memory and demand paging will be discussed.	The students will learn the concept of virtual memory and demand paging.	Peer to peer discussing.	
		Memory Management (Virtual memory concept)	T-1		In this lecture the concept of virtual memory and demand paging will be discussed.	The students will learn the concept of virtual memory and demand paging.	Peer to peer discussing.	
	Lecture 32	Memory Management(Page replacement algorithms)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture simple and multi-level segmentation along with page replacement algorithms will be discussed.	The students will learn simple and multi-level segmentation along with page replacement algorithms	peer to peer discussion and learning by problem solving	

Week 11	Lecture 32	Memory Management (Segmentation - simple, multi-level and with paging)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture simple and multi-level segmentation along with page replacement algorithms will be discussed.	The students will learn simple and multi-level segmentation along with page replacement algorithms	peer to peer discussion and learning by problem solving	
		Memory Management (Fragmentation - internal and external)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture simple and multi-level segmentation along with page replacement algorithms will be discussed.	The students will learn simple and multi-level segmentation along with page replacement algorithms	peer to peer discussion and learning by problem solving	
		Memory Management (Demand paging)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture simple and multi-level segmentation along with page replacement algorithms will be discussed.	The students will learn simple and multi-level segmentation along with page replacement algorithms	peer to peer discussion and learning by problem solving	
	Lecture 33	File Management(File Concepts)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture the concepts of file and directory management will be discussed along with file sharing and mounting concepts will be discussed.	The students will learn simple and multi-level segmentation along with page replacement algorithms.	peer to peer discussion	
		File Management(Access methods)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture the concepts of file and directory management will be discussed along with file sharing and mounting concepts will be discussed.	The students will learn simple and multi-level segmentation along with page replacement algorithms.	peer to peer discussion	
		File Management(Directory Structure)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture the concepts of file and directory management will be discussed along with file sharing and mounting concepts will be discussed.	The students will learn simple and multi-level segmentation along with page replacement algorithms.	peer to peer discussion	
		File Management(File System Mounting and Sharing)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture the concepts of file and directory management will be discussed along with file sharing and mounting concepts will be discussed.	The students will learn simple and multi-level segmentation along with page replacement algorithms.	peer to peer discussion	

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Week 11	Lecture 33	File Management(Directory Implementation)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture the concepts of file and directory management will be discussed along with file sharing and mounting concepts will be discussed.	The students will learn simple and multi-level segmentation along with page replacement algorithms.	peer to peer discussion	
Week 12	Lecture 34	File Management (Protection)	T-1	OR-3 OR-4 OR-6	In this lecture file allocation methods, protection mechanisms and free space management schemes will be discussed.	The students will learn file allocation methods, protection mechanisms and free space management schemes.	Peer to peer discussion and learning by problem solving.	
	Lecture 35	File Management(Allocation methods)	T-1	OR-3 OR-4 OR-6	In this lecture file allocation methods, protection mechanisms and free space management schemes will be discussed.	The students will learn file allocation methods, protection mechanisms and free space management schemes.	Peer to peer discussion and learning by problem solving.	
		File Management(Free-Space Management)	T-1	OR-3 OR-4 OR-6	In this lecture file allocation methods, protection mechanisms and free space management schemes will be discussed.	The students will learn file allocation methods, protection mechanisms and free space management schemes.	Peer to peer discussion and learning by problem solving.	
	Lecture 36	Device management (Dedicated, shared and virtual devices)	T-1	RW-1 RW-2 RW-3	In this lecture different types of devices and access methods will be introduced	The students will learn different types of devices and access methods.	Peer to peer discussion	
		Device management(Serial access and direct access devices)	T-1	RW-1 RW-2 RW-3	In this lecture different types of devices and access methods will be introduced	The students will learn different types of devices and access methods.	Peer to peer discussion	
		Device management(Direct Access Storage Devices – Channels and Control Units)	T-1	RW-1 RW-2 RW-3	In this lecture different types of devices and access methods will be introduced	The students will learn different types of devices and access methods.	Peer to peer discussion	
Week 13	Lecture 37	Device management(Disk scheduling methods)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture various disk scheduling algorithms like FCFS, SSTF, SCAN, C-SCAN, LOOK and C-LOOK will be discussed.	The students will learn various disk scheduling algorithms like FCFS, SSTF, SCAN, C-SCAN, LOOK and C-LOOK.	Learning by problem solving	

Week 13	Lecture 38	Inter process communication (Introduction to IPC (Inter process communication) Methods)	T-1	RW-1 RW-2 RW-3	In this lecture inter process communication concepts and semaphore and mutex will be discussed.	The students will get to know the importance of inter process communication and it's implementation techniques.	peer to peer discussion	
	Lecture 39	Inter process communication (Pipes - popen and pclose functions)	T-1		In this lecture the concept of pipes will be discussed.	the students will be able to understand pipes and their implementation.	peer to peer discussion and learning by doing.	
		Inter process communication (FIFOs)	T-1		In this lecture the concept of pipes will be discussed.	the students will be able to understand pipes and their implementation.	peer to peer discussion and learning by doing.	
Week 14	Lecture 40	Inter process communication (Shared memory)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture shared memory and message passing techniques will be discussed.	the students will learn shared memory and message passing techniques.	peer to peer discussion and learning by problem solving.	
		Inter process communication (Message queues)	T-1	OR-7 RW-1 RW-2 RW-3	In this lecture shared memory and message passing techniques will be discussed.	the students will learn shared memory and message passing techniques.	peer to peer discussion and learning by problem solving.	
		SPILL OVER						
Week 14	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 15	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
	Lecture 45				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0203 (2 best out of 3)

Component	Weightage (%)	Mapped CO(s)
Recoup CA		
Project	50	
Test	50	

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Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Project	The objective of assigning the projects to the students is multi-faceted, aiming to provide them with a holistic understanding of OS working and real-world software development processes.	This CA will be a group project but submission will be on an individual basis.	Individual	Online	30	2 / 10
Test	To assess the students' understanding on Introduction to Operating System, Process Management, CPU Scheduling, and Process Synchronization	This will be an online/offline Quiz/MCQ of 30 Questions with a negative marking of 0.25 marks for every wrong answer based on Introduction to Operating System, Process Management, CPU Scheduling, and Process Synchronization.	Individual	Offline	30	4 / 5
Recoup CA	subjective test based on the syllabus	Deactivated 10June 25Additional chance of CA	Individual	Online	30	8 / 10